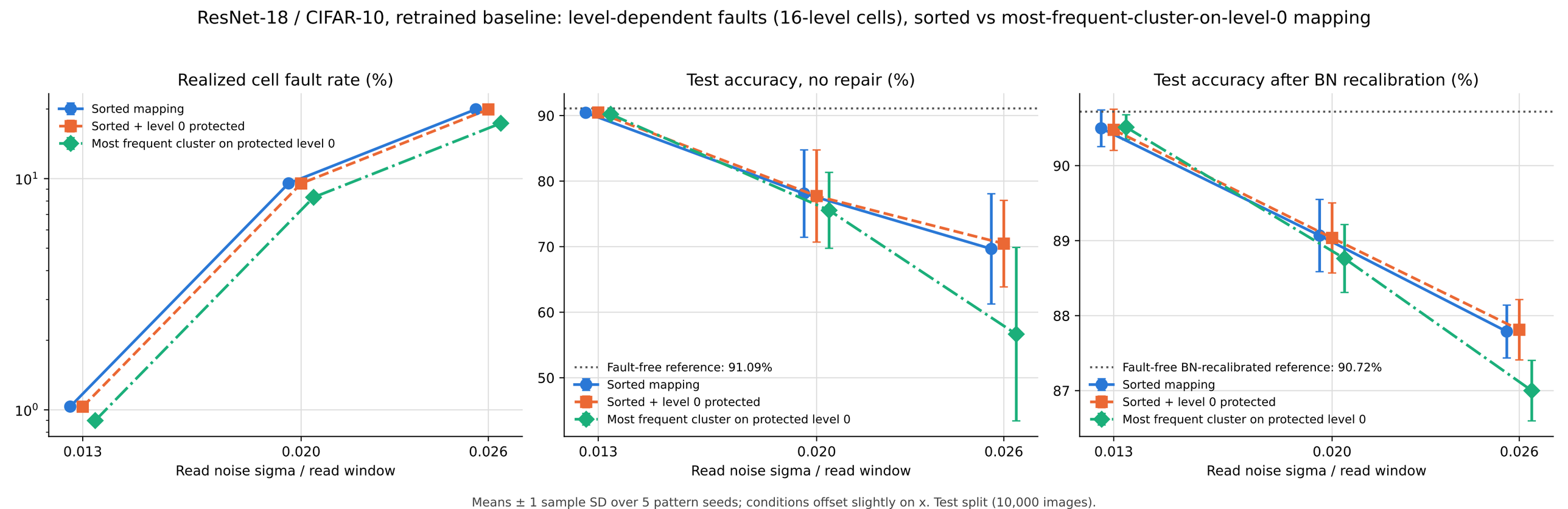
ResNet-18 / CIFAR-10, retrained baseline: level-dependent faults (16-level cells), sorted vs most-frequent-cluster-on-level-0 mapping
Realized cell fault rate (%)
Sorted mapping
Sorted + level 0 protected
Most frequent cluster on protected level 0
10^1
10^0
0.013
0.020
0.026
Read noise sigma / read window
Test accuracy, no repair (%)
90
80
70
60
50
Fault-free reference: 91.09%
Sorted mapping
Sorted + level 0 protected
Most frequent cluster on protected level 0
0.013
0.020
0.026
Read noise sigma / read window
Test accuracy after BN recalibration (%)
90
89
88
87
Fault-free BN-recalibrated reference: 90.72%
Sorted mapping
Sorted + level 0 protected
Most frequent cluster on protected level 0
0.013
0.020
0.026
Read noise sigma / read window
Means ± 1 sample SD over 5 pattern seeds; conditions offset slightly on x. Test split (10,000 images).